

This homework has the goal of sizing the two boosters of Ariane 5 launcher choosing the best material and structural hypothesis to guarantee an acceptable *Margin Of Safety* (MOS) and a structural mass as similar as possible to the real one.

For simplicity it will be considered the evaluation of the size of one of the two boosters. Moreover gravitational and inertial loads of the central body of the launcher are applied to the top of the booster.

DATA

- $\phi = 3100$ mm (booster's diameter)
- $h = 25$ m (booster's height)
- $W_{booster} = 249$ t (booster's weight)
where (as seen from fig. 2):
 - $W_{propellant} = 240$ t
 - $W_{inert\ mass} = 9$ t
- $p = 6.4 \frac{N}{mm^2}$ (pressure which insists on one booster)
- $W_{Ariane\ 5} = 777$ t (Total mass of ARIANE 5)

AXIAL LOAD PRODUCED BY GRAVITATION AND INERTIA

Hp: aerodynamic resistance negligible.

From the load factor evolution sketch in the user's manual of ARIANE 5 it can be considered the first curve because it represents the phase when only the two boosters are working. The maximum load factor is $n_{max} = 4.2$, as it can be seen from fig.3, but it will be used for computations a critical load factor of **n=4.0** which is the value of the load factor when almost all the propellant of the two boosters is consumed.

Now it is evaluated the axial load produced by gravitation and inertia of the central body of the launcher, remembering that we are considering only one booster so the



Figure 1:
Ariane 5

SOLID ROCKET BOOSTER (EAP)	
Size	Ø 3.05 m x 31.6 m
Structure	Stainless steel case
Propulsion	Solid propellant motor (MPS)
Propellants	240 t of solid propellant per EAP
Mean thrust	7000 kN (Vacuum)
Isp	274.5 s
Combustion time	130 s
Attitude control	Steerable nozzle
Avionics	Flight control, flight termination and telemetry subsystems, connected to VEB via data bus + autonomous telemetry

Figure 2: Characteristics of the boosters of ARIANE 5 launcher from ARIANE 5 user's manual

load experienced by it it's half of the total load produced by the central body.
From the given data it can be evaluated the mass of the central body:

$$W_{CB} = W_{Ariane\ 5} - 2W_{booster} = 279\ t$$

So the gravitational force of the central body that insists on one booster is:

$$F = \frac{1}{2}ngW_{CB} = 5473.98\ kN \quad \text{where } g \text{ is the gravitational acceleration}$$

The load's distribution per unit length along one booster is:

$$P = ng \frac{W_{booster}}{h} = 390830\ N/m$$

At the end it can be evaluated the overall axial load experienced by one booster as the sum of the gravitational load due to the central body and the load's distribution of the booster:

$$N_A = F + P * h = 15244.74\ kN \quad (1)$$

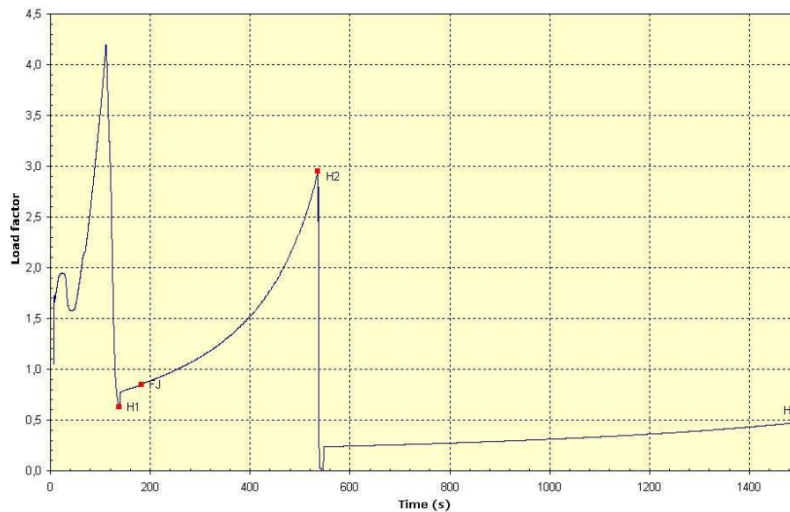


Figure 3: Load factor of ARIANE 5 from ARIANE 5 user's manual.

DESIGN 1

To analyze all the loads that acts on the booster we need to make **hypothesis** on the material, thickness and section of the structure:

- For all the iterative process it will be considered a cylindrical thin walled section which is the real section of ARIANE 5 boosters as it can be seen in fig.1
- Material: **TH1051 (steel)** with $\sigma_y = 965\ Mpa$ and $\rho = 7.80 * 10^3\ Kg/m^3$

- Thickness: $t = 4 \text{ mm}$
- FOS=1.25 from ARIANE 5 user's manual

Now it can be evaluated all the needed factors to have a preliminary sizing of the booster in terms of MOS and structural mass:

AREA: $A = \pi * \phi * t = 0.038936 \text{ m}^2$

ADMISSIBLE LOAD: $N = \sigma_y A = 37573.24 \text{ kN}$

Finally, the MOS is:

$$MOS = \frac{N_{adm}}{FOS N_{applied}} - 1 = \frac{N}{FOS N_A} - 1 = 0.97 \text{ (97\%)}$$

where in this case it was considered as admissible load that due to the yielding stress of the material and as the applied load that evaluated in the previous section due to all the types of loads exerted on the booster.

The structural mass of a cylindrical shape is:

$$W_{str} = (Ah + \pi \frac{\phi^2}{4} 2t) * \rho = 8063.26 \text{ Kg}$$

So considering only these types of loads the structure experiences a very large margin of safety which guarantees that the structure is well designed to experience them. The structural mass turns out to be less than the real one of 9 tons but it is due to the fact that we are not considering the pressurization loads which are fundamental.

Evaluation of the pressurization loads

RADIAL STRESS: $\sigma_R = \frac{pr}{t} = 2.48 \text{ GPa}$

where $r = \frac{\phi}{2} = 1.55 \text{ m}$ and $p = 6.4 \text{ N/mm}^2 = 6.4 * 10^6 \text{ N/m}^2$

LONGITUDINAL STRESS: $\sigma_L = \frac{pr}{2t} = 1.24 \text{ GPa}$

After these first computations it's worth noting that the pressurization loads are higher than the gravitation loads. Moreover, from the theoretical point of view, the gravitational load exerts a compression on the walls of the booster while the pressurization load exerts a traction so, if we want to analyze both of them, we have to consider their different mathematical sign. For simplicity we will consider only the higher load due to the pressurization to design the booster.

Applying the Von-Mises criterion to evaluate the stress related to the design pressurization :

$$\sigma_{VM} = \sqrt{\sigma_R^2 - \sigma_R \sigma_L + \sigma_L^2} = 0.97 \text{ GPa}$$

it's immediately possible to notice that the condition $\sigma_y > \sigma_{VM}$ is not satisfied and the applied stress exceeds the admissible stress due to the material. Moreover in this case the

MOS is:

$$MOS = \frac{\sigma_y}{FOS * \sigma_{VM}} - 1 = -0.204 < 1$$

and when it is considered the pressurization load our first design of the booster is not acceptable so we have to iterate the process considering different hypothesis on the material and structure's thickness.

DESIGN 2

Let's try to change only the thickness of the structure maintaining the same material as the previous one, so:

- Material: **TH1051 (steel)** with $\sigma_y = 965 \text{ Mpa}$ and $\rho = 7.80 * 10^3 \text{ Kg/m}^3$
- Thickness: **$t = 10 \text{ mm}$**

Proceeding as the same way to compute all the needed factors they are obtained:

RADIAL STRESS: $\sigma_R = 0.992 \text{ GPa}$

LONGITUDINAL STRESS: $\sigma_L = 0.496 \text{ GPa}$

then $\sigma_{VM} = 0.859 \text{ GPa}$ and $MOS = -0.101$

From the results it can be seen that even if the condition on the applied stress is satisfied, and in fact $\sigma_y > \sigma_{VM}$, the MOS is equally negative. It means that it is necessary to consider a material with a yielding stress quite higher than the last one.

DESIGN 3

In this case it is maintained the thickness of the second design but it is chosen another type of steel:

- Material: **PH15-7 MO (steel)** with $\sigma_y = 1171 \text{ Mpa}$ and $\rho = 7.60 * 10^3 \text{ Kg/m}^3$
- Thickness: **$t = 10 \text{ mm}$**

Since the thickness is the same of the previous case only the Margin Of Safety changes:

$$MOS = 0.091 = 9.1\%$$

which is an acceptable value but it is mandatory to evaluate also the associated structural mass to see how much this design is well designed in comparison with the real data in fig. 2.

$$A = 0.098596 \text{ m}^2 \quad \text{and} \quad W_{str} = 19880 \text{ Kg}$$

which is very high respect to $W_{inert \text{ mass}}$. This means that it is needed a more performing material which allows to reach a lower structural mass with at the same time a good MOS.

DESIGN 4

Doing a comparative research on *MatWeb* site it is chosen a steel with an higher yielding stress which allows to use a lower thickness:

- Material: **AISI GRADE MARAGING STEEL** with $\sigma_y = 1693 \text{ Mpa}$ and $\rho = 8.0 * 10^3 \text{ Kg/m}^3$
- Thickness: **$t = 7 \text{ mm}$**

Results:

RADIAL STRESS: $\sigma_R = 1.42 \text{ GPa}$

LONGITUDINAL STRESS: $\sigma_L = 0.71 \text{ GPa}$

$\sigma_{VM} = 1.23 \text{ GPa}$ and $MOS = 0.10 = 10\%$

$W_{str} = 14472.51 \text{ Kg}$

The iterative process is going on the right way because it is obtained a good MOS and at the same time the structural mass is almost 5 tons lower than the design 4.

DESIGN 5

Searching for a steel with better yielding stress in order to decrease the thickness and structural mass it is chosen:

- Material: **AISI GRADE MARAGING STEEL** annealed at 815° for 30 min with $\sigma_y = 2164 \text{ Mpa}$ and $\rho = 8.0 * 10^3 \text{ Kg/m}^3$
- Thickness: **$t = 6 \text{ mm}$**

Results:

RADIAL STRESS: $\sigma_R = 1.65 \text{ GPa}$

LONGITUDINAL STRESS: $\sigma_L = 0.825 \text{ GPa}$

$\sigma_{VM} = 1.43 \text{ GPa}$ and $MOS = 0.21 = 21\%$

$W_{str} = 12405 \text{ Kg}$

Thanks to this material it can be able to obtain very interesting results with a Margin Of Safety absolutely satisfying and a structural mass which is the lowest one computed up to this point.

Considering only the axial load due to the gravitational force to verify the structure's performance:

AXIAL LOAD: $N = \sigma_y A = 126386.256 \text{ kN}$ where $A = 0.058404 \text{ m}^2$

MOS: $MOS = \frac{N}{FOS * N_A} - 1 = 5.63$ where N_A is the applied load from the eq.1.

This value allows the structure to work in a large margin of safety.

DESIGN 6

Even if the last design is good from a general point of view, let's try to design a structure taking in consideration a material different from steel which can lead to a good, perhaps better, performances. It is chosen a particular titanium and aluminum alloy with low density but not so high yielding stress:

- Material: **Ti6 Al-4V** with $\sigma_y = 999 \text{ Mpa}$ and $\rho = 4.43 * 10^3 \text{ Kg/m}^3$

After an iterative process the best thickness in according to an admissible MOS (at least 10%) is:

- Thickness: **$t = 12 \text{ mm}$**

Results:

RADIAL STRESS: $\sigma_R = 0.827 \text{ GPa}$

LONGITUDINAL STRESS: $\sigma_L = 0.4135 \text{ GPa}$

$\sigma_{VM} = 0.716 \text{ GPa}$ and $MOS = 0.116 = 11.6\%$

$W_{str} = 13739 \text{ Kg}$

It can be notice that despite the lower density, it is necessary to improve the thickness in order to obtain an admissible MOS and this leads to obtain an higher structural mass than the previous case.

CONCLUSIONS

The best option is the combination between the maraging steel and thickness of 6 mm given by the iteration number 5 as it can be seen in tab.1 where the structural mass is calculated only for design with admissible value of MOS. Thanks to these results it is possible to have a satisfying margin of safety (21%) and a structural mass which can guarantee safe performances from the moment when the two boosters are turned on until all the fuel inside them is consumed. Despite the good results in terms of safety, the best design has a structural mass which is not so close to the theoretical one (9 tons) and this would lead to an higher cost of production. This is probably due to the many simplifications done during the procedure and also the difference between the given data and the real characteristics of ARIANE 5 booster showed in fig. 2.

t [mm]	σ_R [GPa]	σ_L [GPa]	σ_{VM} [GPa]	MOS	W_{str} [Kg]
Material: PH15-7 MO					
$\sigma_y = 1171 \text{ MPa}$ $\rho = 7.60 * 10^3 \text{ Kg/m}^3$					
10	0.992	0.496	0.859	0.091	19880
Material: AISI GRADE MARAGING STEEL					
$\sigma_y = 1693 \text{ MPa}$ $\rho = 8.0 * 10^3 \text{ Kg/m}^3$					
7	1.42	0.71	1.23	0.10	14473
8	1.24	0.62	1.07	0.266	16540
Material: AISI 4053 STEEL					
$\sigma_y = 1538 \text{ MPa}$ $\rho = 7.75 * 10^3 \text{ Kg/m}^3$					
7	1.42	0.71	1.23	0.00033	
8	1.24	0.62	1.07	0.15	16023
Material: AISI GRADE MARAGING STEEL					
$\sigma_y = 2164 \text{ MPa}$ $\rho = 8.0 * 10^3 \text{ Kg/m}^3$					
5	1.98	0.99	1.71	0.012	
6	1.65	0.825	1.43	0.21	12405
Material: Ti6 Al-4V					
$\sigma_y = 999 \text{ MPa}$ $\rho = 4.43 * 10^3 \text{ Kg/m}^3$					
10	0.992	0.496	0.859	-0.070	
11	0.902	0.451	0.781	0.023	
12	0.827	0.4135	0.716	0.116	13739

Table 1: Main results for different materials.